

CSE498 Lab 2: ReVA Video Question Answering

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Project repository: https://github.com/lllingshen/CSE498_lab2

1 Setup and implementation

This lab compares Qwen3-VL-4B-Instruct before and after LoRA fine-tuning, with VILA1.5-3b as a second baseline. I used the course starter code¹ and the official ReVA annotations.² The training subset contains the first 200 training questions. All three models were evaluated on the same 4,000 test questions from 1,014 videos.

The annotation converter produces a conversation for each training question: the user turn contains a video reference, the question, and its lettered options; the assistant turn contains the correct letter inside an `<answer>` tag. Test preparation retains each question's QA ID, video reference, options, answer, and subcategory. The Qwen pipeline evaluates the base model, trains a LoRA adapter, and evaluates that adapter on the same test set. VILA uses its own video-question-answering adapter and is evaluated without fine-tuning.

I completed the core assignment functions in these files:

- `scripts/build_qwen_train_data.py`: converts annotations into video-question conversations and answer targets.
- `reva_eval/data/rsvidqa/prepare_reva_v2_test_set.py`: prepares test questions with stable QA IDs and video references.
- `scripts/score_reva_predictions.py`: parses Qwen answers and scores every ground-truth question.
- `vila_eval/reva_v2.py`: loads ReVA questions, builds VILA prompts, parses answers, and summarizes results.

Other local execution changes include `scripts/run_eval_qwen_finetuned.sh`, `reva_eval/inference_vllm_origin_number.py`, and `scripts/check_student_setup.py`. These check LoRA adapter loading, handle video metadata, and support the local environment and data paths. Single-GPU SDPA training support is implemented in `qwen_finetune/qwenvl/train/train_qwen.py` and `qwen_finetune/scripts/sft_7b.sh`. Focused regression checks are in `tests/test_local_fixes.py`.

The saved training arguments record one epoch, batch size 1, gradient accumulation of 4, and learning rate 2×10^{-7} . LoRA uses rank 8, alpha 16, and dropout 0 on the attention query, key, value, and output projections. The training state records 50 optimizer steps and mean training loss 0.4115. The saved evaluation configuration points to the trained adapter and the Qwen base model; the run log records adapter loading followed by successful model and processor loading.

¹https://github.com/likaiw2/VLM_evaluating_finetuning

²<https://huggingface.co/datasets/ReVA-Benchmark/ReVA>

The recorded experiments use a maximum of four frames per video and greedy decoding. Qwen receives a request for reasoning followed by an answer tag, with a maximum of 1,024 output tokens. VILA receives a request for only the option letter, with a maximum of 128 output tokens. These are the settings used for the saved runs, rather than recommendations for further tuning. Reproduction commands are in `README.md`.

Scores were recomputed from the saved raw responses. The saved evaluation ground truth matches the official test annotations in QA IDs, questions, options, labels, video references, and subcategories. Each run has all 4,000 unique IDs and no duplicates. The scoring rule is

$$\text{Accuracy} = \frac{\text{number of correct answers}}{\text{all ground-truth questions}}.$$

Missing or unparseable answers count as wrong. Completion means that an answer can be parsed into an option letter; it does not mean the answer is correct. An explicit final answer takes precedence over distractors mentioned in the explanation.

2 Results and interpretation

Model	Correct / total	Accuracy	Parseable / total
Qwen base	2,675 / 4,000	66.88%	3,971 / 4,000
Qwen LoRA	2,673 / 4,000	66.83%	3,964 / 4,000
VILA base	1,940 / 4,000	48.50%	4,000 / 4,000

LoRA did not improve overall accuracy in this experiment. It produced two fewer correct answers, a decrease of 0.05 percentage points. A direct question-by-question comparison shows that 93 previously wrong answers became correct, while 95 correct answers became wrong. All models generated responses for all questions, but 29 base-Qwen responses and 36 LoRA responses were unparseable and therefore counted as incorrect.

There were some differences by subcategory. In Geometric Relation, Qwen improved from 270/400 (67.50%) to 280/400 (70.00%) after fine-tuning. Temporal Grounding remained difficult: the base model answered 252/640 (39.38%) correctly, compared with 250/640 (39.06%) after fine-tuning. These local changes did not yield an overall improvement. The training loss and these scores alone do not establish overfitting or the cause of the difference. Qwen outperformed VILA under the settings used here, with the prompt and output-budget differences noted above.

3 Three qualitative examples

1. Counting cars: QA-000005 (visual perception). The question asks how many cars are visible in the parking lot at 00:00. The options are A: 40–50, B: 10–20, C: 20–30, and D: 30–40. The official answer is **B**; Qwen base predicts **D**, Qwen LoRA predicts **C**, and VILA predicts **B**. Both Qwen responses select a count range above the official answer. The LoRA response changes the range but remains incorrect. This is consistent with a visual counting difficulty, although the outputs alone do not show why the models differ. The timestamp matters: the question asks for the count at the beginning, not a count accumulated across the video.

2. Visibility interval: QA-000010 (temporal reasoning). The question asks how long a group of people remains visible in the parking lot. The options are A: 1–5 s, B: 0–4 s, C: 0–5 s, and D: 2–5 s. The official answer is **C**; Qwen base and Qwen LoRA both predict **C**, while VILA predicts **D**. VILA selects a later starting time and a shorter duration than the official answer. The options differ by only one or two seconds, making this a useful temporal-reasoning example. With only four sampled frames, the input provides limited evidence for precise boundaries. Choosing the correct interval does not by itself demonstrate continuous tracking of the group.

3. Viewpoint and answer mapping: QA-000248 (option parsing). The question asks for the camera viewpoint. The options are A: high angle ($> 45^\circ$), B: top-down (90°), C: high angle ($< 45^\circ$), and D: ground-level (0°). The official answer is **A**; Qwen base predicts **C**, Qwen LoRA predicts **A**, and VILA predicts **B**. The base model’s explanation includes “high angle (> 45 degrees)”, but its final tag is `<answer>C</answer>`. This is a model answer-mapping inconsistency, not a scorer error: the final letter must be scored as C even though the explanation describes option A. The LoRA response aligns its explanation with A. This one corrected example does not establish an overall benefit from fine-tuning.

4 Limitations and conclusion

This was a small, one-epoch training run, and the four-frame budget limits the available video information. The models also use different prompts and output limits. The 200 training questions come from 17 videos; 16 of those videos also occur in the test set, covering 68 test questions. No identical video/question pair occurs in both this training subset and the test set, but the evaluation is not entirely on unseen videos. These limitations do not establish the cause of any score change.

The experiment shows how to convert video annotations, train and load a LoRA adapter, and compare models using a common test denominator. Qwen performed better than VILA in this run, while the small LoRA fine-tuning run left Qwen’s overall accuracy essentially unchanged.